

Statistical Optimization

Lagrange duality

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Constrained convex programming

- Convex programs:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & && h_i(x) = 0, \quad i = 1, \dots, p \end{aligned} \tag{1}$$

where $f_0, \dots, f_m$ are convex, and $h_1, \dots, h_p$ are affine.

- Feasible set

$$X = \{x \in \mathbb{R}^d : f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p\},$$

which is convex.

- Domain

$$\mathcal{D} = \left(\bigcap_{i=0}^m \text{dom}(f_i) \right) \cap \left(\bigcap_{i=1}^p \text{dom}(h_i) \right),$$

often simply taken as $\mathbb{R}^d$.

Outline

Lagrange Duality

KKT Conditions

Applications of KKT

Lagrange duality

Definition. Given optimization problem (1), its *Lagrangian* is the function $L : \mathcal{D} \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ defined by

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x). \quad (2)$$

The λ_i, ν_i are called *Lagrange multipliers*.

The *Lagrange dual function* is the function $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R} \cup \{-\infty\}$ defined by

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu). \quad (3)$$

Lagrange duality

Theorem (Weak Lagrange duality). Let x be a feasible solution of the optimization problem (1), meaning that $f_i(x) \leq 0$ for $i = 1, \dots, m$ and $h_i(x) = 0$ for $i = 1, \dots, p$. Let g be the Lagrange dual function of (1) and $\lambda \in \mathbb{R}^m, \nu \in \mathbb{R}^p$ such that $\lambda \geq 0$. Then

$$g(\lambda, \nu) \leq f_0(x).$$

Proof. For any x belonging to the feasible set,

$$g(\lambda, \nu) \leq L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

$$\underbrace{\sum_{i=1}^m \lambda_i f_i(x)}_{\leq 0} + \underbrace{\sum_{i=1}^p \nu_i h_i(x)}_{=0} \Rightarrow g(\lambda, \nu) \leq f_0(x).$$

Lagrange Dual Problem

Definition. Let g be the Lagrange dual function of the optimization problem (1). Then the Lagrange dual of (1) is the optimization problem

$$\begin{aligned} & \text{maximize} && g(\lambda, \nu) \\ & \text{subject to} && \lambda \geq 0. \end{aligned} \tag{4}$$

- The equivalent minimization problem

$$\begin{aligned} & \text{minimize} && -g(\lambda, \nu) \\ & \text{subject to} && \lambda \geq 0 \end{aligned}$$

is a convex program, even if (1) is not.

Strong Duality

Theorem. Suppose that the convex program (1) has a feasible solution x^* that additionally satisfies $f_i(x^*) < 0$, $i = 1, \dots, m$ (a Slater point). Then

$$\inf_{x \in X} f_0(x) = \sup_{\lambda \geq 0, \nu \in \mathbb{R}^p} g(\lambda, \nu)$$

for its Lagrange dual (4). Moreover, if this value is finite, it is attained by a feasible solution of the dual (4).

- Strong duality may still hold even without a Slater point, or even when (1) is not convex. Slater's condition simply provides one powerful sufficient condition.

Zero duality gap

Definition (Zero duality gap). Let x^* be feasible for the primal (1) and (λ^*, ν^*) feasible for the Lagrange dual (4). The primal and dual solutions x^* and (λ^*, ν^*) are said to have *zero duality gap* if

$$f_0(x^*) = g(\lambda^*, \nu^*).$$

Zero Duality Gap and Its Consequences

If x^* and (λ^*, ν^*) have zero duality gap, then we have the chain of (in)equalities:

$$\begin{aligned} f_0(x^*) &= g(\lambda^*, \nu^*) \\ &= \inf_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x) \right) \\ &\leq f_0(x^*) + \underbrace{\sum_{i=1}^m \lambda_i^* f_i(x^*)}_{\leq 0} + \underbrace{\sum_{i=1}^p \nu_i^* h_i(x^*)}_{=0} \\ &\leq f_0(x^*). \end{aligned} \tag{5}$$

All inequalities must be equalities.

Consequences from Zero Duality Gap (KKT)

Lemma (Complementary slackness). If zero duality gap holds, then

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m.$$

Lemma (Vanishing Lagrangian gradient). If f_i and h_i are differentiable,

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^p \nu_i^* \nabla h_i(x^*) = 0.$$

Minimax Formulation of Lagrangian Duality

For any $x \in \mathcal{D}$,

$$\sup_{\lambda \geq 0, \nu \in \mathbb{R}^p} L(x, \lambda, \nu) = \begin{cases} f_0(x), & x \text{ feasible,} \\ +\infty, & x \text{ infeasible.} \end{cases}$$

Therefore, the primal problem can be written as

$$\min_{x \in X} f_0(x) = \min_{x \in \mathcal{D}} \sup_{\lambda \geq 0, \nu \in \mathbb{R}^p} L(x, \lambda, \nu).$$

By exchanging the order of optimization, we also obtain the dual problem

$$\sup_{\lambda \geq 0, \nu \in \mathbb{R}^p} \inf_{x \in \mathcal{D}} L(x, \lambda, \nu) = \sup_{\lambda \geq 0, \nu \in \mathbb{R}^p} g(\lambda, \nu).$$

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KKT Conditions

Consider a primal feasible point x^* for (1) and a dual feasible pair (λ^*, ν^*) for (4). The **KKT conditions** are:

Primal feasibility: $f_i(x^*) \leq 0, \quad i = 1, \dots, m, \quad h_j(x^*) = 0, \quad j = 1, \dots, p,$

Dual feasibility: $\lambda_i^* \geq 0, \quad i = 1, \dots, m,$

Complementary slackness: $\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m,$

Stationarity:
$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0.$$

Two ways:

zero duality gap and local minimum + constraint qualification.

KKT necessary conditions (zero duality gap)

Theorem. Let x^* be feasible for the primal problem (1), and let (λ^*, ν^*) be feasible for the dual problem (4). Assume that all f_i and h_i are differentiable.

If

$$f_0(x^*) = g(\lambda^*, \nu^*),$$

then (x^*, λ^*, ν^*) satisfies the KKT conditions:

Primal feasibility: $f_i(x^*) \leq 0, \quad i = 1, \dots, m, \quad h_j(x^*) = 0, \quad j = 1, \dots, p,$

Dual feasibility: $\lambda_i^* \geq 0, \quad i = 1, \dots, m,$

Complementary slackness: $\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m,$

Stationarity:
$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0.$$

KKT necessary conditions (local minimum)

Theorem. Assume that all f_i and h_j in (1) are differentiable. Let x^* be a local minimum of (1) satisfying a constraint qualification (e.g., LICQ: the gradients of the active inequality constraints and the equality constraints at x^* are linearly independent). Then there exist $\lambda^* \in \mathbb{R}^m$ and $\nu^* \in \mathbb{R}^p$ such that

Primal feasibility: $f_i(x^*) \leq 0, \quad i = 1, \dots, m, \quad h_j(x^*) = 0, \quad j = 1, \dots, p,$

Dual feasibility: $\lambda_i^* \geq 0, \quad i = 1, \dots, m,$

Complementary slackness: $\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m,$

Stationarity:
$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0.$$

Proof (Optional)

Proof. Let

$$I(x^*) = \{i \in \{1, \dots, m\} : f_i(x^*) = 0\}$$

be the active set.

Since x^* is a local minimum of (1), it is feasible:

$$f_i(x^*) \leq 0, \quad h_j(x^*) = 0.$$

If there existed $d \in \mathbb{R}^d$ such that

$$\nabla f_0(x^*)^\top d < 0, \quad \nabla f_i(x^*)^\top d < 0 \quad (i \in I(x^*)), \quad \nabla h_j(x^*)^\top d = 0 \quad (j = 1, \dots, p),$$

then $x^* + td$ would be feasible for all sufficiently small $t > 0$ and

$$f_0(x^* + td) < f_0(x^*),$$

contradicting the local optimality of x^* .

Proof (Optional)

Hence no such d exists. By Farkas' lemma (a theorem of alternatives), there exist $\lambda_i^* \geq 0$ for $i \in I(x^*)$ and $\nu_j^* \in \mathbb{R}$ such that

$$\nabla f_0(x^*) + \sum_{i \in I(x^*)} \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0.$$

Set $\lambda_i^* = 0$ for $i \notin I(x^*)$. Then

$$\lambda_i^* \geq 0, \quad \lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m.$$

Together with feasibility of x^* , this gives the four KKT conditions.



KKT sufficient conditions

Theorem (KKT sufficient conditions). Let x^* and (λ^*, ν^*) be feasible solutions of the primal optimization problem (1) and its Lagrange dual (4), respectively. Further suppose that all f_i and h_i in (1) are differentiable, all f_i are **convex**, and all h_i are **affine**, and that

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m, \quad (6)$$

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^p \nu_i^* \nabla h_i(x^*) = 0. \quad (7)$$

Then x^* is a **global minimum** of the primal problem (1), and the primal and dual problems have **zero duality gap**.

Proof (1/3)

Proof. Let x be any feasible point. By convexity of f_0 ,

$$f_0(x) \geq f_0(x^*) + \nabla f_0(x^*)^\top (x - x^*).$$

Using (7),

$$\nabla f_0(x^*) = - \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) - \sum_{i=1}^p \nu_i^* \nabla h_i(x^*),$$

so

$$f_0(x) - f_0(x^*) \geq - \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*)^\top (x - x^*) - \sum_{i=1}^p \nu_i^* \nabla h_i(x^*)^\top (x - x^*).$$

Proof (2/3)

By convexity of each f_i ,

$$f_i(x) \geq f_i(x^*) + \nabla f_i(x^*)^\top (x - x^*),$$

hence

$$\nabla f_i(x^*)^\top (x - x^*) \leq f_i(x) - f_i(x^*).$$

Since each h_i is affine,

$$\nabla h_i(x^*)^\top (x - x^*) = h_i(x) - h_i(x^*).$$

Therefore,

$$f_0(x) - f_0(x^*) \geq - \sum_{i=1}^m \lambda_i^* (f_i(x) - f_i(x^*)) - \sum_{i=1}^p \nu_i^* (h_i(x) - h_i(x^*)).$$

Proof (3/3)

Since x and x^* are feasible,

$$f_i(x) \leq 0, \quad h_i(x) = 0, \quad h_i(x^*) = 0.$$

Also, by dual feasibility and complementary slackness,

$$\lambda_i^* \geq 0, \quad \lambda_i^* f_i(x^*) = 0.$$

Hence

$$\begin{aligned} f_0(x) - f_0(x^*) &\geq - \sum_{i=1}^m \lambda_i^* (f_i(x) - f_i(x^*)) \\ &= - \sum_{i=1}^m \lambda_i^* f_i(x) + 0 \geq 0. \end{aligned}$$

Thus $f_0(x) \geq f_0(x^*)$ for every feasible x , so x^* is primal optimal. By weak duality, (λ^*, ν^*) is dual optimal as well.

Global optimality, zero duality gap, and KKT

Relationship map.

- **Weak duality** always holds:

$$g(\lambda, \nu) \leq f_0(x).$$

- **Global optimality** $\Rightarrow$ **KKT** requires a constraint qualification (e.g. LICQ); convexity is not needed.
- **KKT** $\Rightarrow$ **global optimality** requires convexity: $f_0, \dots, f_m$ convex and $h_1, \dots, h_p$ affine.
- Under **convexity + strong duality**, the following are equivalent:

$$\text{global optimality} \iff \text{zero duality gap} \iff \text{KKT}.$$

- Without convexity, the implications are only one-way: KKT need not imply global optimality, and zero duality gap need not imply KKT.

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Applications of KKT

Example 1: projection onto the ℓ_1 ball

Consider the problem

$$\min_{\theta \in \mathbb{R}^d} \frac{1}{2} \|\theta - \eta\|_2^2 \quad \text{s.t.} \quad \|\theta\|_1 \leq \tau.$$

This is exactly the projection step in projected gradient descent for constrained Lasso:

$$\Pi_{\|\theta\|_1 \leq \tau}(\eta) = \arg \min_{\|\theta\|_1 \leq \tau} \frac{1}{2} \|\theta - \eta\|_2^2.$$

We first reformulate this problem into a smooth constrained optimization problem, and then apply the standard KKT conditions.

KKT conditions and the soft-thresholding form

We first claim that at the optimum, each coordinate has the same sign as η_i :

$$\theta_i^* \eta_i \geq 0, \quad i = 1, \dots, d.$$

Indeed, changing the sign of θ_i to match η_i does not change $\|\theta\|_1$, but can only decrease $(\theta_i - \eta_i)^2$.

So we may write

$$\theta_i^* = \text{sign}(\eta_i) z_i^*, \quad z_i^* \geq 0.$$

Then

$$|\theta_i^*| = z_i^*, \quad (\theta_i^* - \eta_i)^2 = (z_i^* - |\eta_i|)^2.$$

Hence the projection problem becomes

$$\min_{z \in \mathbb{R}^d} \frac{1}{2} \sum_{i=1}^d (z_i - |\eta_i|)^2 \quad \text{s.t.} \quad \sum_{i=1}^d z_i \leq \tau, \quad z_i \geq 0.$$

KKT conditions and the soft-thresholding form

Now the problem is smooth, so we apply the standard KKT conditions. Its Lagrangian is

$$\mathcal{L}(z, \lambda, \nu) = \frac{1}{2} \sum_{i=1}^d (z_i - |\eta_i|)^2 + \lambda \left(\sum_{i=1}^d z_i - \tau \right) - \sum_{i=1}^d \nu_i z_i,$$

where $\lambda \geq 0, \nu_i \geq 0$. The vanishing Lagrangian gradient condition is

$$z_i^* - |\eta_i| + \lambda^* - \nu_i^* = 0.$$

Together with complementary slackness

$$\nu_i^* z_i^* = 0,$$

we obtain

$$z_i^* = \max(|\eta_i| - \lambda^*, 0).$$

Therefore

$$\theta_i^* = \text{sign}(\eta_i) z_i^* = \text{sign}(\eta_i) \max(|\eta_i| - \lambda^*, 0).$$

How do we determine λ^* ?

There are two cases.

Case 1: If $\|\eta\|_1 \leq \tau$, then η is already feasible, so

$$\theta^* = \eta, \quad \lambda^* = 0.$$

Case 2: If $\|\eta\|_1 > \tau$, then the constraint is active, so by complementary slackness

$$\sum_{i=1}^d z_i^* = \tau.$$

Using

$$z_i^* = \max(|\eta_i| - \lambda^*, 0),$$

we get

$$\sum_{i=1}^d \max(|\eta_i| - \lambda^*, 0) = \tau.$$

Example 2: Simplex projection

We consider the problem

$$\begin{aligned} \min_{\theta \in \mathbb{R}^n} \quad & \frac{1}{2} \|\theta - \eta\|^2 \\ \text{s.t.} \quad & \sum_{i=1}^n \theta_i - 1 = 0, \\ & -\theta_i \leq 0, \quad i = 1, \dots, n. \end{aligned}$$

The Lagrangian is

$$L(\theta, \lambda, \nu) = \frac{1}{2} \|\theta - \eta\|^2 + \nu \left(\sum_{i=1}^n \theta_i - 1 \right) - \sum_{i=1}^n \lambda_i \theta_i.$$

Simplex projection: KKT conditions

The KKT conditions are:

1. Stationarity:

$$\nabla_{\theta_i} L = \theta_i - \eta_i + \nu - \lambda_i = 0, \quad i = 1, \dots, n.$$

2. Primal feasibility:

$$\sum_{i=1}^n \theta_i = 1, \quad \theta_i \geq 0.$$

3. Dual feasibility:

$$\lambda_i \geq 0.$$

4. Complementary slackness:

$$\lambda_i \theta_i = 0.$$

Simplex projection: solving the solution

From stationarity,

$$\theta_i = \eta_i - \nu + \lambda_i.$$

Now use complementary slackness:

$$\theta_i > 0 \implies \lambda_i = 0 \implies \theta_i = \eta_i - \nu.$$

$$\theta_i = 0 \implies \lambda_i = \nu - \eta_i \geq 0 \implies \eta_i \leq \nu.$$

Therefore,

$$\theta_i = (\eta_i - \nu)_+, \quad i = 1, \dots, n.$$

The equality constraint then gives

$$\sum_{i=1}^n (\eta_i - \nu)_+ = 1.$$

Simplex projection: proposition and remark

Let

$$\Delta^n := \left\{ \theta \in \mathbb{R}^n : \theta_i \geq 0, \sum_{i=1}^n \theta_i = 1 \right\}.$$

Proposition. The Euclidean projection of η onto Δ^n is

$$\theta_i^* = (\eta_i - \tau)_+, \quad i = 1, \dots, n,$$

where τ is the unique scalar satisfying

$$\sum_{i=1}^n (\eta_i - \tau)_+ = 1.$$

Remark. The solution is sparse. Complementary slackness tells us $\theta_i = 0$ whenever $\eta_i < \tau$ without any computation.

KL divergence

For probability vectors on Δ^n , a natural non-Euclidean distance-like quantity is KL divergence

$$D_{\text{KL}}(\theta \parallel \eta) := \sum_{i=1}^n \theta_i \log \frac{\theta_i}{\eta_i}.$$

Remark. It can be derived as the Bregman divergence associated with the negative entropy

$$\Phi(\theta) := \sum_{i=1}^n \theta_i \log \theta_i.$$

Intuition. It measures how different two distributions are, and is more suitable than the Euclidean distance when θ represents probabilities.

Example 3: Mirror descent on the simplex

In Example 2, we used the Euclidean distance

$$\frac{1}{2} \|\theta - \eta\|^2.$$

Now we use the KL divergence as the proximity term. Later, we will show that the resulting update is equivalent to an optimization problem involving

$$\sum_{i=1}^n \theta_i \log \theta_i.$$

We study the problem

$$\begin{aligned} \min_{\theta \in \mathbb{R}^n} \quad & \sum_{i=1}^n \theta_i \log \theta_i - \langle \theta, \eta \rangle \\ \text{s.t.} \quad & \sum_{i=1}^n \theta_i - 1 = 0. \end{aligned}$$

This is the entropic analogue of simplex projection.

Entropy regularization: Lagrangian and stationarity

The Lagrangian is

$$L(\theta, \nu) = \sum_{i=1}^n \theta_i \log \theta_i - \langle \theta, \eta \rangle + \nu \left(\sum_{i=1}^n \theta_i - 1 \right).$$

The stationarity condition is

$$\nabla_{\theta_i} L = \log \theta_i + 1 - \eta_i + \nu = 0, \quad i = 1, \dots, n.$$

Hence

$$\theta_i = e^{\eta_i - 1 - \nu}, \quad i = 1, \dots, n.$$

Entropy regularization: softmax solution

Using the constraint

$$\sum_{i=1}^n \theta_i = 1,$$

we obtain

$$\theta_i = \frac{e^{\eta_i}}{\sum_{j=1}^n e^{\eta_j}}, \quad i = 1, \dots, n.$$

Therefore,

$$\theta^* = \text{softmax}(\eta).$$

So, under entropy geometry, the closed-form solution is no longer thresholding; it becomes a normalized exponential map.

Mirror descent on simplex

We want to minimize f over the probability simplex Δ^n . At the current iterate θ_t , let

$$g_t := \nabla f(\theta_t).$$

Intuition. On the simplex, Euclidean geometry is not the most natural one. Instead of taking an additive gradient step and projecting in Euclidean norm, mirror descent uses the KL geometry of probability distributions.

It can be viewed in three steps:

$$\theta_t \longrightarrow \log \theta_t \longrightarrow \log \theta_t - \gamma_t g_t \longrightarrow \theta_{t+1}.$$

Equivalently, θ_{t+1} is obtained by the non-Euclidean projection

$$\theta_{t+1} = \arg \min_{\theta \in \Delta^n} \left\{ \langle g_t, \theta \rangle + \frac{1}{\gamma_t} D_{\text{KL}}(\theta \| \theta_t) \right\},$$

where

$$D_{\text{KL}}(\theta \| \theta_t) = \sum_{i=1}^n \theta_i \log \frac{\theta_i}{(\theta_t)_i}.$$

Equivalent form

We start from the mirror descent update

$$\theta_{t+1} = \arg \min_{\theta \in \Delta^n} \left\{ \langle \mathbf{g}_t, \theta \rangle + \frac{1}{\gamma_t} \sum_{i=1}^n \theta_i \log \frac{\theta_i}{(\theta_t)_i} \right\}.$$

Expand the logarithm and the objective becomes

$$\langle \mathbf{g}_t, \theta \rangle + \frac{1}{\gamma_t} \sum_{i=1}^n \theta_i \log \theta_i - \frac{1}{\gamma_t} \langle \theta, \log \theta_t \rangle.$$

Multiplying by $\gamma_t > 0$ does not change the $\arg \min$, so this is equivalent to

$$\theta_{t+1} = \arg \min_{\theta \in \Delta^n} \left\{ \sum_{i=1}^n \theta_i \log \theta_i - \langle \theta, \eta_t \rangle \right\}.$$

where

$$\eta_t := \log \theta_t - \gamma_t \mathbf{g}_t.$$

Mirror descent update on Δ^n

Applying the previous softmax formula with $\eta = \eta_t$, we get

$$\theta_{t+1} = \text{softmax}(\eta_t).$$

Equivalently,

$$(\theta_{t+1})_i = \frac{(\theta_t)_i e^{-\gamma_t(g_t)_i}}{\sum_{j=1}^n (\theta_t)_j e^{-\gamma_t(g_t)_j}}, \quad i = 1, \dots, n.$$

Remark. These two examples illustrate the power of the Lagrangian formulation: it often leads to many closed-form update rules.